

L5-1.5-Solving Quadratic by Factoring

MCR3U

Review of Radicals

Simplify each of the following:

1) $(\sqrt{2} + 3\sqrt{3})(5\sqrt{3} - 10)$

2)
$$\frac{2 - \sqrt{80}}{4}$$

3) $4\sqrt{10}(3 + 2\sqrt{2})$

How to Solve Quadratics

Solving a quadratic means to find the x-intercepts (or roots).

To solve a quadratic equation:

- 1) It must be set to equal 0. Before factoring, it must be in the form $ax^2 + bx + c = 0$
- 2) Factor the left side of the equation
- 3) Set each factor to equal zero and solve for 'x'.

zero product rule: if two factors have a product of zero; one or both of the factors must equal zero.

Example 1: Solve the following quadratics by factoring

a) $0 = x^2 - 15x + 56$

When factoring $ax^2+bx+c=0$ when 'a' is 1 or can be factored out

Steps to follow:

- 1) Check if there is a common factor that can be divided out
- 2) Look at the 'c' value and the 'b' value
- 3) Determine what factors multiply to give 'c' and add to give 'b'
- 4) put those factors into $(x+r)(x+s)$ for 'r' and 's'.
- 5) make sure nothing else can be factored

b) $-6 = x^2 - 5x$

c) $0 = 2x^2 - 8x - 42$

Example 2: Solve by factoring

a) $8x^2 + 2x - 15 = 0$

Steps for factoring $ax^2 + bx + c$ when $a \neq 1$

- 1) Check for any common factors that can be factored out
- 2) Replace the middle term bx with two terms whose coefficients have a sum of b and a product of $a \times c$
- 3) Group pairs of terms and remove a common factor from each pair
- 4) Remove the common binomial factor

b) $2x^2 - 11x = -15$

Example 3: For the quadratic $y = 2x^2 - 4x - 16$

a) Find the roots of the quadratic by factoring

b) Find the axis of symmetry (average of x-intercepts)

c) Find the coordinates of the vertex and state if it is a max or min value